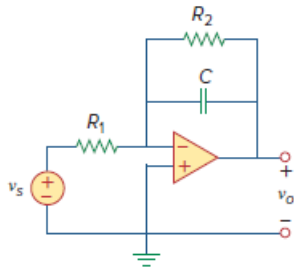


Problem

Using Fig. 10.113, design a problem to help other students better understand op amps in AC circuits.

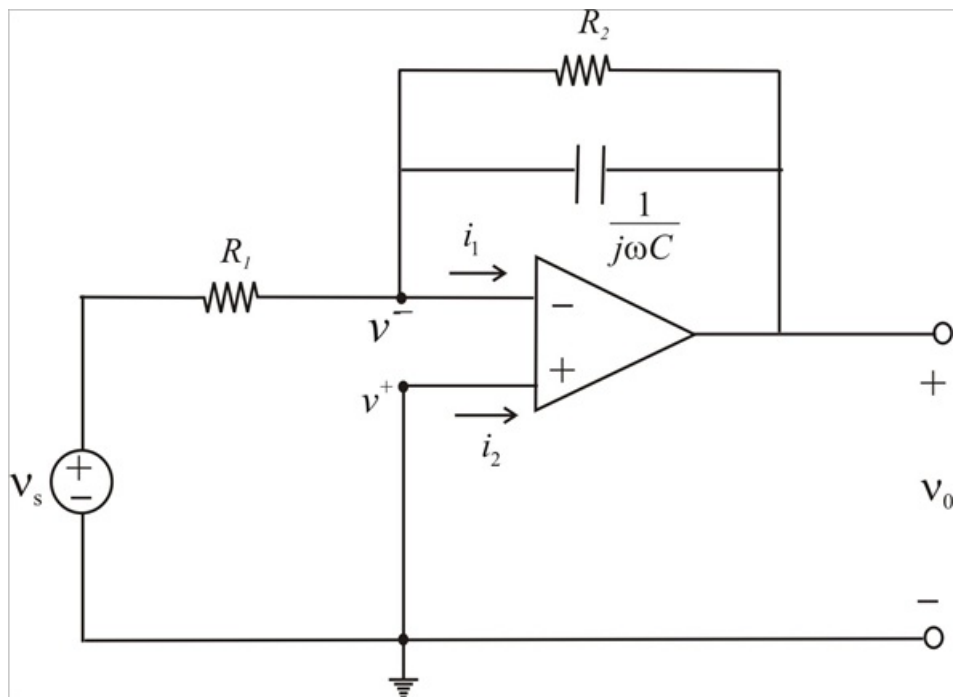
Figure 10.113:



Step-by-step solution

Step 1 of 3

Given circuit can be represented as,



Step 2 of 3

From the above circuit.

$$v^+ = 0V$$

Currents through inverting and non – inverting terminals of an ideal OP amp are zero.

i.e,

$$i_1 = i_2 = 0A$$

The voltages at inverting and non – inverting nodes of an ideal OP amp are equal.

i.e,

$$v^- = v^+$$

$$\therefore v^- = v^+ = 0V$$

Step 3 of 3

Applying KCL at node v^-

$$\frac{v^- - v_s}{R_1} + \frac{v^- - v_0}{R_2} + \frac{v^- - v_0}{\frac{1}{j\omega C}} = 0$$

$$\frac{(0 - v_s)}{R_1} + \left(\frac{0 - v_0}{R_2} \right) + \left(\frac{(0 - v_0)j\omega C}{1} \right) = 0$$

$$\frac{-v_s R_2 - v_0 R_1 - v_0 j\omega C R_1 R_2}{R_1 R_2} = 0$$

$$-(R_1 + j\omega C R_1 R_2) v_0 = v_s R_2$$

$$\therefore v_0 = -v_s \left[\frac{R_2}{R_1 + j\omega C R_1 R_2} \right]$$